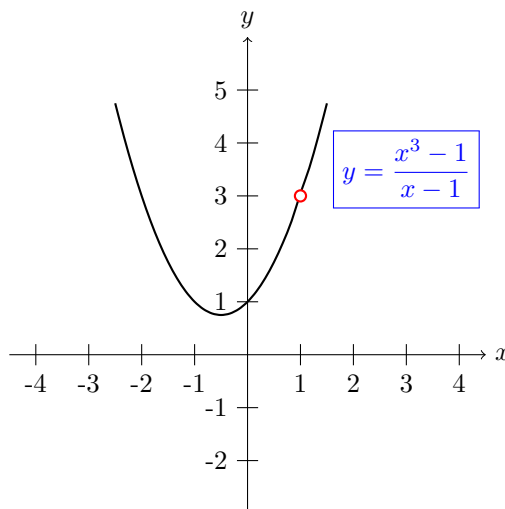


Lecture Two

1 Limit of the Function

1.1 Introduction

Given the function $f(x) = \frac{x^3 - 1}{x - 1}$, clearly it is not defined at $x = 1$, and we want to study the behaviour of this function when x approaches to 1.



The following table shows:

x	0.75	0.9	0.99	0.999
$f(x)$	2.313	2.710	2.970	2.997

From the left

x	1.001	1.01	1.1	1.25
$f(x)$	3.003	3.030	3.310	3.813

From the right

We can move arbitrarily close to 1 and as a result $f(x)$ moves arbitrarily close to 3, and here we can say the limit of the function $f(x) = \frac{x^3 - 1}{x - 1}$ is 3 when x goes to 1. This limit is written as:

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = 3$$

In general: If $f(x)$ becomes arbitrarily close to a single number L as x approaches c from either sides, the limit of $f(x)$ as x approaches c is L , and we can write it as:

$$\lim_{x \rightarrow c} f(x) = L$$

1.2 Definition of limit

Let f be a function defined on an open interval containing c (except possibly at c) and let L be a real number. The statement $\lim_{x \rightarrow c} f(x) = L$ means that for each $\epsilon > 0$ there exist $\delta > 0$ such that if $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

Example 1.2.1 Prove that $\lim_{x \rightarrow 3} (2x + 3) = 9$.

Proof

We must show that for each $\epsilon > 0$, there exists a $\delta > 0$ such that $|(2x+3)-9| < \epsilon$ whenever $0 < |x-3| < \delta$, because our choice of δ depends on ϵ . Now we need to establish a connection between the absolute value $|(2x + 3) - 9|$ and $|x - 3|$.

$$|(2x + 3) - 9| = |2x - 6| = 2|x - 3|$$

So for a given $\epsilon > 0$, we can choose $\delta = \frac{\epsilon}{2}$. This choice works because

$$0 < |x - 3| < \delta = \frac{\epsilon}{2}$$

implies that

$$|(2x + 3) - 9| = 2|x - 3| < 2\left(\frac{\epsilon}{2}\right) = \epsilon.$$

1.3 One Sided Limit

1. **Left Hand Limit:** We mean by L-H-L, limit of the function when x increasing to c , ($x \neq c$), i.e. ($x < c$) and $x \rightarrow c$ and we denote it by

$$\lim_{x \rightarrow c^-} f(x)$$

To evaluate the left limit, we substitute $x = c - \epsilon$ on $f(x)$, then we take limit when $\epsilon \rightarrow 0$. ($\epsilon > 0$)

$$\lim_{x \rightarrow c^-} f(x) = \lim_{\epsilon \rightarrow 0} f(c - \epsilon)$$

2. **Right Hand Limit:** We mean by R-H-L, limit of the function when x decreasing to c , ($x \neq c$), i.e. ($x > c$) and $x \rightarrow c$ and we denote it by

$$\lim_{x \rightarrow c^+} f(x)$$

To evaluate the left limit, we substitute $x = c + \epsilon$ on $f(x)$, then we take limit when $\epsilon \rightarrow 0$. ($\epsilon > 0$)

$$\lim_{x \rightarrow c^+} f(x) = \lim_{\epsilon \rightarrow 0} f(c + \epsilon)$$

And the limit of the function **will be exists** if and only if:

$$\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x)$$

Example 1.3.2 Find the limit $\lim_{x \rightarrow 0} \frac{|x|}{x}$ using L-H-L and R-H-L

Sol

1. L-H-L:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{|x|}{x} &= \lim_{\epsilon \rightarrow 0} \frac{|0 - \epsilon|}{0 - \epsilon} = \frac{|-\epsilon|}{-\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon}{-\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} -1 = -1. \end{aligned}$$

2. R-H-L:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{|x|}{x} &= \lim_{\epsilon \rightarrow 0} \frac{|0 + \epsilon|}{0 + \epsilon} = \frac{|\epsilon|}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} 1 = 1. \end{aligned}$$

Since $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$, therefore the limit does not exist.

Example 1.3.3 Find the limit $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x - 1}$ using L-H-L and R-H-L

Sol

1. L-H-L $\lim_{x \rightarrow 1^-} \frac{x^2 + x - 2}{x - 1}$

$$\begin{aligned} \lim_{x \rightarrow 1^-} \frac{x^2 + x - 2}{x - 1} &= \lim_{\epsilon \rightarrow 0} \frac{(1 - \epsilon)^2 + (1 - \epsilon) - 2}{(1 - \epsilon) - 1} = \lim_{\epsilon \rightarrow 0} \frac{1 - 2\epsilon + \epsilon^2 + 1 - \epsilon - 2}{-\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon(\epsilon - 3)}{-\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} (3 - \epsilon) = 3 - 0 = 3 \end{aligned}$$

2. R-H-L: $\lim_{x \rightarrow 1^+} \frac{x^2 + x - 2}{x - 1}$

$$\begin{aligned} \lim_{x \rightarrow 1^+} \frac{x^2 + x - 2}{x - 1} &= \lim_{\epsilon \rightarrow 0} \frac{(1 + \epsilon)^2 + (1 + \epsilon) - 2}{(1 + \epsilon) - 1} = \lim_{\epsilon \rightarrow 0} \frac{1 + 2\epsilon + \epsilon^2 + 1 + \epsilon - 2}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon(\epsilon + 3)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} (\epsilon + 3) = 0 + 3 = 3 \end{aligned}$$

Since R-H-L = L-H-L, the limit exists and equal to 3.

1.4 Properties of Limits

Theorem 1.4.1 Let b and c be real numbers and let n be a positive integer.

1.

$$\lim_{x \rightarrow c} b = b$$

2.

$$\lim_{x \rightarrow c} x = c$$

3.

$$\lim_{x \rightarrow c} x^n = c^n$$

Example 1.4.4 Evaluate the limits

1.

$$\lim_{x \rightarrow 2} 3 = 3$$

2.

$$\lim_{x \rightarrow -4} x = -4$$

3.

$$\lim_{x \rightarrow 2} x^2 = (2)^2 = 4$$

Theorem 1.4.2 Let b and c be real numbers and let n be a positive integer, and let f and g be functions with the following limits.

$$\lim_{x \rightarrow c} f(x) = L \quad , \quad \lim_{x \rightarrow c} g(x) = K$$

then

1. Scalar multiple

$$\lim_{x \rightarrow c} [bf(x)] = b \lim_{x \rightarrow c} f(x) = bL$$

2. Sum or Difference

$$\lim_{x \rightarrow c} [f(x) \pm g(x)] = \lim_{x \rightarrow c} f(x) \pm \lim_{x \rightarrow c} g(x) = L \pm K$$

3. Product

$$\lim_{x \rightarrow c} [f(x).g(x)] = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x) = L.K$$

4. Quotient

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} = \frac{L}{K} \quad , K \neq 0$$

5. Power

$$\lim_{x \rightarrow c} [f(x)]^n = \left[\lim_{x \rightarrow c} f(x) \right]^n = L^n.$$

Example 1.4.5 By using theorems of limits evaluate the following limits.

1. $\lim_{x \rightarrow 2} (4x^2 + 3)$

Sol

$$\begin{aligned}\lim_{x \rightarrow 2} (4x^2 + 3) &= \lim_{x \rightarrow 2} 4x^2 + \lim_{x \rightarrow 2} 3 = 4 \lim_{x \rightarrow 2} x^2 + \lim_{x \rightarrow 2} 3 \\ &= 4 \left(\lim_{x \rightarrow 2} x \right)^2 + \lim_{x \rightarrow 2} 3 = 4(2)^2 + 3 = 19.\end{aligned}$$

$$2. \lim_{x \rightarrow 3} \left[\frac{x-2}{x+2} \right]$$

Sol

$$\lim_{x \rightarrow 3} \left[\frac{x-2}{x+2} \right] = \frac{\lim_{x \rightarrow 3} (x-2)}{\lim_{x \rightarrow 3} (x+2)} = \frac{\lim_{x \rightarrow 3} x - \lim_{x \rightarrow 3} 2}{\lim_{x \rightarrow 3} x + \lim_{x \rightarrow 3} 2} = \frac{3-2}{3+2} = \frac{1}{5}$$

$$3. \lim_{x \rightarrow 4} \sqrt{25 - x^2}$$

Sol

$$\lim_{x \rightarrow 4} (25 - x^2)^{\frac{1}{2}} = \left(\lim_{x \rightarrow 4} 25 - \lim_{x \rightarrow 4} x^2 \right)^{\frac{1}{2}} = \left(\lim_{x \rightarrow 4} 25 - \left(\lim_{x \rightarrow 4} x \right)^2 \right)^{\frac{1}{2}} = (25 - (4)^2)^{\frac{1}{2}} = 3$$

$$4. \lim_{x \rightarrow 1} (3x^2 - 1)(9x + 2)$$

Example 1.4.6 Evaluate the following limits:

$$1. \lim_{x \rightarrow 1} (x + 4 - x^2)$$

Sol

$$2. \lim_{x \rightarrow 3} \frac{3x+1}{x+2}$$

Sol

$$3. \lim_{x \rightarrow 1} (x+4)(x-1)$$

Sol

$$4. \lim_{x \rightarrow 2} \frac{x+2}{x-2}$$

Sol

$$5. \lim_{x \rightarrow \infty} \frac{3 + \frac{1}{x}}{2 + x}$$

Sol

$$6. \lim_{x \rightarrow \infty} \frac{4x + 1}{x + 9}$$

Sol

$$7. \lim_{x \rightarrow -1} \frac{x^2 - 1}{x + 1}$$

Sol

$$8. \lim_{x \rightarrow -1} \frac{x^2 + 3x + 2}{x + 1}$$

Sol

$$9. \lim_{x \rightarrow 2} \frac{x^5 - 32}{x^2 - 4}$$

Sol

$$10. \lim_{x \rightarrow 0} \frac{x + \sin x}{\tan x + x}$$

Sol

1.5 Strategies for Finding Limits

Previously, we studied several types of functions whose limits can be evaluated by **direct substitution**, this knowledge together with the following theorem can be used to develop a strategy for finding limits.

Theorem 1.5.3 Let c be a real number and let $f(x) = g(x)$ for all $x \neq c$ in an open interval containing c . If the limit of $g(x)$ as x approaches c exists, then the limit of $f(x)$ also exists and:

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x)$$

Example 1.5.7 Evaluate the following limits:

1. $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 4}$

Sol

By direct substitution, we get:

$$\frac{(2)^2 - 2 - 2}{(2)^2 - 4} = \frac{4 - 2 - 2}{4 - 4} = \frac{0}{0} \quad (\text{undetermined value})$$

Now we try to find a function g that agrees with f for all x other than $x = 2$. By factoring and dividing out like factors, we can rewrite the given function as:

$$f(x) = \frac{x^2 - x - 2}{x^2 - 4} = \frac{\cancel{(x-2)}(x+1)}{\cancel{(x-2)}(x+2)} = \frac{x+1}{x+2} = g(x).$$

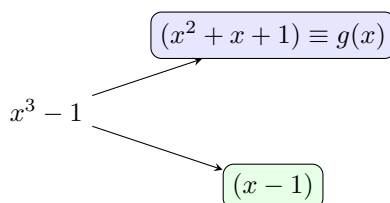
So for all x -values other than $x = 2$ the functions f and g agree, since $\lim_{x \rightarrow 2} g(x)$ exists, we can apply theorem 1.5.3 to conclude that f and g have the same limit at $x = 2$.

$$\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{x+1}{x+2} = \frac{2+1}{2+2} = \frac{3}{4}$$

2. $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$

Sol

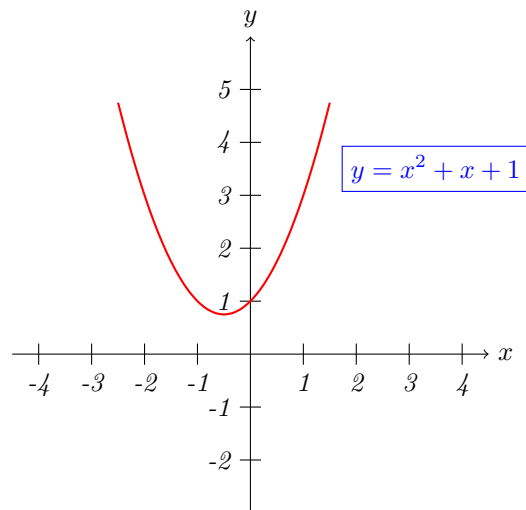
By direct substitution we get $\frac{(1)^3 - 1}{1 - 1} = \frac{0}{0}$. Then we try to find a function g that agrees with f for all x other than $x = 1$.



$$f(x) = \frac{x^3 - 1}{x - 1} = \frac{(x-1)(x^2 + x + 1)}{(x-1)} = (x^2 + x + 1) = g(x)$$

Thus $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} (x^2 + x + 1) = 1^2 + 1 + 1 = 3$.

Note that: the graph of the function $g(x) = x^2 + x + 1$ is same as $f(x) = \frac{x^3 - 1}{x - 1}$



3. $\lim_{x \rightarrow -1} \frac{x+1}{x^2+4x+3}$

Sol

$$\lim_{x \rightarrow -1} \frac{x+1}{x^2+4x+3} = \lim_{x \rightarrow -1} \frac{x+1}{(x+1)(x+3)} = \lim_{x \rightarrow -1} \frac{1}{(x+3)} = \frac{1}{-1+3} = \frac{1}{2}$$

4. $\lim_{x \rightarrow 5} \frac{x-5}{\sqrt{x-3}-\sqrt{2}}$

Sol

$$\lim_{x \rightarrow 5} \frac{x-5}{\sqrt{x-3}-\sqrt{2}} \cdot \frac{\sqrt{x-3}+\sqrt{2}}{\sqrt{x-3}+\sqrt{2}} = \lim_{x \rightarrow 5} \frac{(x-5)(\sqrt{x-3}+\sqrt{2})}{(x-5)} = \lim_{x \rightarrow 5} (\sqrt{x-3}+\sqrt{2}) = \sqrt{2}+\sqrt{2} = 2\sqrt{2}.$$

5. $\lim_{x \rightarrow 0} \frac{(1+x)^2-1}{x}$

Sol

6. $\lim_{x \rightarrow 1} \frac{x^6-1}{x^2-1}$

Sol

7. $\lim_{x \rightarrow 3} \frac{\frac{1}{x} - \frac{1}{3}}{x^2-9}$

Sol

$$8. \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$$

Sol

Example 1.5.8 Evaluate the following limits.

$$1. \lim_{x \rightarrow \infty} \frac{9x^2}{2x^2 + 4}$$

Sol

By direct substitution, we get:

$$\lim_{x \rightarrow \infty} \frac{9x^2}{2x^2 + 4} = \frac{9(\infty)^2}{2(\infty)^2 + 4} = \frac{\infty}{\infty}, (\text{undetermined value})$$

$$\lim_{x \rightarrow \infty} \underbrace{\frac{9x^2}{2x^2 + 4}}_{f(x)} = \lim_{x \rightarrow \infty} \left(\frac{\cancel{x^2}[9]}{\cancel{x^2}\left[2 + \frac{4}{x^2}\right]} \right) = \lim_{x \rightarrow \infty} \underbrace{\left(\frac{9}{2 + \frac{4}{x^2}} \right)}_{g(x)} = \frac{9}{2 + \frac{4}{\infty^2}} = \frac{9}{2 + 0} = \frac{9}{2}$$

$$2. \lim_{x \rightarrow \infty} \frac{x^3 - x^2 + 2}{2x^2 + x + 1}$$

Sol

$$\lim_{x \rightarrow \infty} \underbrace{\frac{x^3 - x^2 + 2}{2x^2 + x + 1}}_{f(x)} = \lim_{x \rightarrow \infty} \frac{\cancel{x^3}\left(1 - \frac{1}{x} + \frac{2}{x^3}\right)}{\cancel{x^3}\left(\frac{2}{x} + \frac{1}{x^2} + \frac{1}{x^3}\right)} = \lim_{x \rightarrow \infty} \frac{\left(1 - \frac{1}{x} + \frac{2}{x^3}\right)}{\underbrace{\left(\frac{2}{x} + \frac{1}{x^2} + \frac{1}{x^3}\right)}_{g(x)}} = \frac{\left(1 - \frac{1}{\infty} + \frac{2}{\infty^3}\right)}{\left(\frac{2}{\infty} + \frac{1}{\infty^2} + \frac{1}{\infty^3}\right)} = \infty$$

$$3. \lim_{x \rightarrow \infty} \frac{x(x+1)}{x^3+1}$$

Sol

$$4. \lim_{x \rightarrow \infty} \frac{6x + 1}{\sqrt{4x^2 + 3}}$$

Sol

$$5. \lim_{x \rightarrow \infty} \frac{(x - 3)(2x + 1)}{2x^2 - x + 1}$$

Sol

$$6. \lim_{x \rightarrow \infty} \frac{(3x + 4)(x - 2)}{x(2x + 1)(x + 2)}$$

Sol

$$7. \lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 5}}{x + 3}$$

Sol

1.6 Cauchy's Theorem

Theorem 1.6.4 *Cauchy's Theorem:*

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

proof

By using **long division**, we get:

$$\frac{x^n - a^n}{x - a} = x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-1}$$

Therefore:

$$\begin{aligned}\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} &= \lim_{x \rightarrow a} (x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-1}) \\ &= a^{n-1} + a \times a^{n-2} + a^2 \times a^{n-3} + \cdots + a^{n-1} \\ &= \underbrace{a^{n-1} + a^{n-1} + a^{n-1} + \cdots + a^{n-1}}_{n \text{ times}} = na^{n-1}\end{aligned}$$

Lemma 1.6.1

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}$$

proof

We can write $f(x) = \frac{x^n - a^n}{x^m - a^m} = \frac{x^n - a^n}{x - a} \div \frac{x^m - a^m}{x - a}$. Then the limit will be:

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} \div \lim_{x \rightarrow a} \frac{x^m - a^m}{x - a} = \frac{na^{n-1}}{ma^{m-1}} = \frac{n}{m} a^{n-m}.$$

Lemma 1.6.2

$$\lim_{x \rightarrow a} \frac{x - a}{x^m - a^m} = \frac{1}{m} a^{1-m}$$

Proof left for you.

Example 1.6.9 Evaluate the following limits.

1. $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^3 - 8}$

Sol

$$\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^3 - 8} = \lim_{x \rightarrow 2} \frac{x^5 - (2)^5}{x^3 - (2)^3} = \frac{5}{3} (2)^{5-3} = \frac{5}{3} (2)^2 = \frac{20}{3}.$$

2. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x\sqrt{x} - 8}$

Sol

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x\sqrt{x} - 8} = \lim_{x \rightarrow 4} \frac{x^2 - (4)^2}{x^{\frac{3}{2}} - (4)^{\frac{3}{2}}} = \frac{2}{\frac{3}{2}} (4)^{2-\frac{3}{2}} = \frac{4}{3} (2) = \frac{8}{3}$$

3. $\lim_{\Delta x \rightarrow 0} \frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x}$

Sol

$$\lim_{\Delta x \rightarrow 0} \frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x} = \lim_{x + \Delta x \rightarrow x} \frac{(x + \Delta x)^{\frac{1}{2}} - x^{\frac{1}{2}}}{(x + \Delta x) - x} = \frac{1}{2} x^{\frac{1}{2}-1} = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

4. $\lim_{x \rightarrow 3} \frac{x - 3}{x^3 - 27}$

Sol

$$\lim_{x \rightarrow 3} \frac{x - 3}{x^3 - 27} = \lim_{x \rightarrow 3} \frac{x - 3}{x^3 - (3)^3} = \frac{1}{3} (3)^{1-3} = \frac{1}{3} \cdot \frac{1}{9} = \frac{1}{27}.$$

5. $\lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1}$

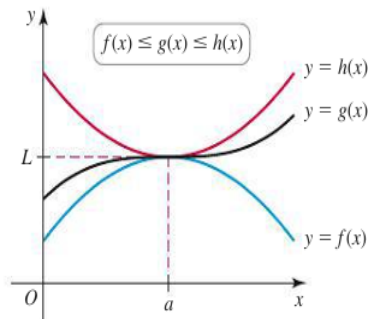
Sol

$$\lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1} = \lim_{x \rightarrow -1} \frac{x^3 - (-1)^3}{x - (-1)} = 3(-1)^{3-1} = 3(-1)^2 = 3.$$

1.7 The squeeze Theorem

Theorem 1.7.5 Assume the functions f, g and h satisfy $f(x) \leq g(x) \leq h(x)$ for all values of x near a , except possibly at a .

$$\boxed{\text{If } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L, \text{ then } \lim_{x \rightarrow a} g(x) = L}$$



1.8 Limit of Trigonometric Functions

We have some important limits:

$$1. \quad \boxed{\lim_{x \rightarrow 0} \sin x = 0}$$

$$2. \quad \boxed{\lim_{x \rightarrow 0} \cos x = 1}$$

$$3. \quad \boxed{\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1}$$

$$4. \quad \boxed{\lim_{ax \rightarrow 0} \frac{\sin ax}{ax} = 1}$$

$$5. \quad \boxed{\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1}$$

$$6. \quad \boxed{\lim_{ax \rightarrow 0} \frac{\tan ax}{ax} = 1}$$

Example 1.8.10 1. Prove that $\boxed{\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1}$

Sol

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x \cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1 \times \frac{1}{\cos 0} = 1 \times 1 = 1$$

$$2. \quad \lim_{x \rightarrow 0} \frac{\sin 2x}{x}$$

Sol

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = \lim_{2x \rightarrow 0} \frac{2 \sin 2x}{2x} = 2 \lim_{2x \rightarrow 0} \frac{\sin 2x}{2x} = 2 \times 1 = 2$$

$$3. \quad \lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$$

Sol

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\tan 3x}{x} = \frac{1}{2} \lim_{3x \rightarrow 0} \frac{3 \tan 3x}{3x} = \frac{3}{2} \lim_{3x \rightarrow 0} \frac{\tan 3x}{3x} = \frac{3}{2} \times 1 = \frac{3}{2}.$$

$$4. \lim_{x \rightarrow 0} \frac{\tan x + \sin 3x}{x}$$

Sol

$$\lim_{x \rightarrow 0} \frac{\tan x + \sin 3x}{x} = \lim_{x \rightarrow 0} \frac{\tan x}{x} + \lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} \frac{\tan x}{x} + 3 \lim_{3x \rightarrow 0} \frac{\sin 3x}{3x} = 1 + 3 \times 1 = 4.$$

$$5. \lim_{x \rightarrow 0} \frac{2x + \sin x}{4x + \tan 5x}$$

Sol

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{2x + \sin x}{4x + \tan 5x} &= \lim_{x \rightarrow 0} \left(\frac{\frac{2x + \sin x}{x}}{\frac{4x + \tan 5x}{x}} \right) = \lim_{x \rightarrow 0} \frac{2x + \sin x}{x} \div \lim_{x \rightarrow 0} \frac{4x + \tan 5x}{x} \\ &= \left(\lim_{x \rightarrow 0} 2 + \lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \div \left(\lim_{x \rightarrow 0} 4 + \lim_{x \rightarrow 0} \frac{\tan 5x}{x} \right) \\ &= \left(\lim_{x \rightarrow 0} 2 + \lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \div \left(\lim_{x \rightarrow 0} 4 + 5 \lim_{5x \rightarrow 0} \frac{\tan 5x}{5x} \right) \\ &= \frac{2+1}{4+5} = \frac{3}{9} = \frac{1}{3}. \end{aligned}$$

$$6. \lim_{x \rightarrow \infty} x \sin \left(\frac{3}{x} \right)$$

Sol

Let $u = \frac{3}{x} \Rightarrow x = \frac{3}{u}$ and $u \rightarrow 0$ as $x \rightarrow \infty$. Thus

$$\lim_{x \rightarrow \infty} x \sin \left(\frac{3}{x} \right) = \lim_{u \rightarrow 0} \frac{3}{u} \sin u = 3 \lim_{u \rightarrow 0} \frac{\sin u}{u} = 3 \times 1 = 3.$$

$$7. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

Sol

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \times \frac{1 + \cos x}{1 + \cos x} &= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x^2(1 + \cos x)} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2(1 + \cos x)} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \times \lim_{x \rightarrow 0} \frac{1}{(1 + \cos x)} \\ &= \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2 \times \lim_{x \rightarrow 0} \frac{1}{(1 + \cos x)} = 1 \times \frac{1}{(1 + \cos 0)} = \frac{1}{2} \end{aligned}$$

1.9 The Number e

We define the number e by the limit

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$$

Proof

$$\begin{aligned} \left(1 + \frac{1}{x} \right)^x &= 1 + \binom{x}{1} \cdot \frac{1}{x} + \binom{x}{2} \cdot \frac{1}{x^2} + \binom{x}{3} \cdot \frac{1}{x^3} + \dots + \binom{x}{n} \cdot \frac{1}{x^n} \\ &= \sum_{n=0}^x \frac{1}{n!} \left(1 - \frac{1}{x} \right) \cdot \left(1 - \frac{2}{x} \right) \cdot \left(1 - \frac{3}{x} \right) \dots \left(1 - \frac{n-1}{x} \right) \end{aligned}$$

Therefore the limit will be:

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x &= \sum_{n=0}^x \left(\lim_{x \rightarrow \infty} \frac{1}{n!} \left(1 - \frac{1}{x} \right) \cdot \left(1 - \frac{2}{x} \right) \cdot \left(1 - \frac{3}{x} \right) \dots \left(1 - \frac{n-1}{x} \right) \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots \end{aligned}$$

The number e satisfied the inequality $2 < e < 3$, and its value ≈ 2.71828 , and we can write the limit as:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = \lim_{n \rightarrow 0} (1 + n)^{\frac{1}{n}}$$

1.10 The Exponential Theorem

Theorem 1.10.6

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Proof

We know that $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

$$\Rightarrow e^x = \left[\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \right]^x = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{nx}$$

put $m = nx \Rightarrow e^x = \lim_{m \rightarrow \infty} \left(1 + \frac{x}{m}\right)^m$, therefore the limit will be:

$$\begin{aligned} &\Rightarrow \lim_{m \rightarrow \infty} \left[1 + \frac{mx}{m} + \frac{m(m-1)}{2!} \frac{x^2}{m^2} + \frac{m(m-1)(m-2)}{3!} \frac{x^3}{m^3} + \cdots \right] \\ &= \lim_{m \rightarrow \infty} \left[1 + x + \frac{\left(1 - \frac{1}{m}\right)}{2!} + \frac{\left(1 - \frac{1}{m}\right)\left(1 - \frac{2}{m}\right)}{3!} + \cdots \right] \\ &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \end{aligned}$$

Example 1.10.11 1. $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$

Sol

put $u = \frac{1}{x} \Rightarrow x = \frac{1}{u}$ and $u \rightarrow \infty$ as $x \rightarrow 0$

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = \lim_{u \rightarrow \infty} \left(1 + \frac{1}{u}\right)^u = e.$$

2. $\lim_{x \rightarrow \infty} \left(1 + \frac{\alpha}{x}\right)^x \quad \alpha \neq 0$

Sol

(i) If $\alpha > 0$, let $x = \alpha u$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{\alpha}{x}\right)^x = \lim_{u \rightarrow \infty} \left(1 + \frac{\alpha}{\alpha u}\right)^{\alpha u} = \left(\lim_{u \rightarrow \infty} \left(1 + \frac{1}{u}\right)^u \right)^{\alpha} = e^{\alpha}.$$

(ii) If $\alpha < 0 \Rightarrow$ let $x = -\alpha u$

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(1 + \frac{\alpha}{x}\right)^x &= \lim_{u \rightarrow \infty} \left(1 - \frac{\alpha}{\alpha u}\right)^{-\alpha u} = \left(\lim_{u \rightarrow \infty} \left(1 - \frac{1}{u}\right)^{-u} \right)^{\alpha} = \left(\lim_{u \rightarrow \infty} \left(\frac{u}{u-1}\right)^u \right)^{\alpha} \\ &= \left[\lim_{u \rightarrow \infty} \left(1 + \frac{1}{u-1}\right)^u \right]^{\alpha} \end{aligned}$$

Now let $u-1 = y \Rightarrow u = 1+y$ ($y \rightarrow \infty$ as $u \rightarrow \infty$).

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(1 + \frac{\alpha}{x}\right)^x &= \left[\lim_{u \rightarrow \infty} \left(1 + \frac{1}{u-1}\right)^u \right]^{\alpha} = \left[\lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^{y+1} \right]^{\alpha} \\ &= \left[\lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^y \times \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right) \right]^{\alpha} = (e \times 1)^{\alpha} = e^{\alpha}. \end{aligned}$$

$$3. \lim_{x \rightarrow \infty} \left(\frac{x+3}{x-2} \right)^x$$

Sol

Let $u = x - 2 \Rightarrow x = u + 2 \Rightarrow (u \rightarrow \infty) \text{ as } (x \rightarrow \infty)$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{x+3}{x-2} \right)^x = \lim_{u \rightarrow \infty} \left(\frac{u+2+3}{u} \right)^{u+2} = \lim_{u \rightarrow \infty} \left(1 + \frac{5}{u} \right)^{u+2} = \lim_{u \rightarrow \infty} \left(1 + \frac{5}{u} \right)^u \cdot \lim_{u \rightarrow \infty} \left(1 + \frac{5}{u} \right)^2 = e^5$$

Exercise 1.10.1 Evaluate the following limits.

$$1. \lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$$

Sol

$$2. \lim_{x \rightarrow 4} \frac{x^2 - 16}{4 - x}$$

Sol

$$3. \lim_{x \rightarrow b} \frac{(x-b)^{50} - x + b}{x - b}$$

Sol

$$4. \lim_{h \rightarrow 0} \frac{\frac{1}{5+h} - \frac{1}{5}}{h}$$

Sol

$$5. \lim_{w \rightarrow 1} \left(\frac{1}{w^2 - w} - \frac{1}{w - 1} \right)$$

Sol

$$6. \lim_{x \rightarrow \infty} \frac{x^2 + 2x + 9}{3x^3 + x + 1}$$

Sol

$$7. \lim_{x \rightarrow 1} \frac{\sqrt{9x^2 + 7}}{3x + 1}$$

Sol

$$8. \lim_{n \rightarrow \infty} \left(\frac{1 + 2 + 3 + \cdots + n}{n^2} \right)$$

Sol

$$9. \lim_{x \rightarrow 1} \frac{x^3 + x^2 + 5x - 5}{x - 1}$$

Sol

$$10. \lim_{x \rightarrow 0} \left(\frac{\frac{1}{x-1} + \frac{1}{x+1}}{x} \right)$$

Sol

$$11. \lim_{x \rightarrow 2} \frac{x^3(x-1)^6 - 8}{x - 2}$$

Sol

$$12. \lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 12} - 4}{x - 2}$$

Sol

$$13. \lim_{x \rightarrow 1} \frac{x^{-1} - 1}{x - 1}$$

Sol

$$14. \lim_{x \rightarrow 4} \frac{4x^2 - x^2}{2 - \sqrt{x}}$$

Sol

$$15. \lim_{x \rightarrow 4} \frac{4 - x}{\sqrt{x^2 + 9} - 5}$$

Sol

$$16. \lim_{y \rightarrow 8} \frac{\sqrt[3]{y} - 2}{y \sqrt[3]{y} - 16}$$

Sol

$$17. \lim_{y \rightarrow 3} \frac{y^4 - 81}{y^2 - 9}$$

Sol

$$18. \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^3 - x^3}{\Delta x}$$

Sol

$$19. \lim_{x \rightarrow -2} \frac{x^5 + 32}{x^3 + 8}$$

Sol

$$20. \lim_{x \rightarrow 0} \frac{4x + \sin 3x}{x}$$

Sol

$$21. \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 9x}$$

Sol

$$22. \lim_{x \rightarrow 0} \frac{\sin 2x}{x^2 + 3x}$$

Sol

$$23. \lim_{x \rightarrow 2} \frac{\sin(x - 2)}{x^2 - 4}$$

Sol

$$24. \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2}$$

Sol

$$25. \lim_{x \rightarrow 0} \frac{\tan 2x + 4x}{x + \sin x + \tan x}$$

Sol

$$26. \lim_{x \rightarrow 0} \frac{\cos 4x - \cos 2x}{x^2}$$

Sol

$$27. \lim_{x \rightarrow \infty} x \sin \frac{6}{x}$$

Sol

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